L02 VISUAL LIGHTING DESIGN | P

Intent : Provide visual comfort and enhance visual acuity for all users through electric lighting.

Summary : This WELL feature requires projects to provide appropriate illuminances on work planes for regular users of all age groups, as required for the tasks performed in the space.

Issue : Humans perceive the world through visual cues that are received through images formed on the retina of the eye. The light levels in a space can enhance the user's ability to perform tasks in that space, while contributing to the feeling of spaciousness. The age of the individual is also a factor in the amount of light required for visual acuity. As humans age, the transmission of light through their lenses is reduced. This is due to age-related changes, including increased light absorption by the lenses, smaller pupil size, increased scattering of light due to thicker lenses and yellowing of the lenses.[1,2] This aging of the eye indicates that an increase in light levels is required to ensure visual acuity.

Solutions : While developing a lighting strategy to accommodate the visual acuity of users, it is critical to take into account the tasks conducted, as well as the age of the users. Projects may refer to published recommendations by lighting associations or authorities on using electric lighting design strategies for light levels required on the work plane. Lighting recommendations published by authorities provide a range of lighting levels for different age groups and tasks.

PART 1 PROVIDE VISUAL ACUITY

For All Spaces except Dwelling Units:

Option 1: Visual lighting design

The following requirements are met:

a. All indoor and outdoor spaces (including transition areas) comply with the illuminance thresholds specified in one of the following lighting reference guidelines:
   1. IES Lighting Library, Lighting Applications Standards Collection.[3]
   2. EN 12464-1:2021[4] or EN 12464-2:2014.[5]
   3. ISO 8995-1:2002(E) (CIE S 008/E:2001).[6]
   4. GB50034-2013.[7]
   5. CIBSE SLL Code for Lighting.[8]

b. The illuminance thresholds take into consideration the tasks and the age groups of the occupants.

OR

Option 2: Predetermined light levels

The following requirements are met:

a. More than 50% of the occupants are under the age of 65.

b. The area of outdoor space within the project boundary is less than 5% of the interior project area.

c. At least 90% of the interior project area is comprised of the following space types and meets the associated illuminance thresholds:

Room Types	Minimum Illuminance Threshold
Offices Classrooms	320 lux at task surface.[9]
Circulation areas (including lobbies and atria) Storage spaces	110 lux at floor level.[9]
Dining areas Lounges Restrooms	110 lux at task surface.[9]

For Dwelling Units:

Option 1: Promote Visual Acuity

The following requirements are met:

a. Lighting is installed in kitchens and bathrooms to comply with the illuminance thresholds specified in one of the following lighting reference guidelines:
   1. IES Lighting Library, Lighting Applications Standards Collection.[3]
   2. ISO 8995-1:2002(E) (CIE S 008/E:2001).[6]
   3. GB50034-2013.[7]
   4. CIBSE SLL Code for Lighting.[8]

b. For spaces where lighting is not installed, the following is provided to all tenants:
    1. Illuminance thresholds for common tasks conducted in spaces
    2. Specifications, quantity and location of light fixtures required to meet light levels based on sample layout

Note :
Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details.

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. Pokorny J, Smith VC, Lutze M. Aging of the human lens. Appl Opt. 1987;26(8):1437. doi:10.1364/AO.26.001437
2. Owsley C. Aging and vision. Vision Res. 2011;51(13):1610-1622. doi:10.1016/j.visres.2010.10.020
3. Illuminating Engineering Society. 2020. IES OL-IM-03: Lighting Applications Standards Collection Subscription. https://store.ies.org/product/lighting-applications-standards-collection-subscription/?v=7516fd43adaa
4. European Committee for Standardization. 2021. EN 12464-1:2021. Light and lighting - Lighting of work places - Part 1: Indoor work places. Accessed Febraury February 13, 2024. https://standards.iteh.ai/catalog/standards/sist/b1a68cbc-1bdc-4ac2-89e7-0123982763b5/sist-en-12464-2-2014
5. European Committee for Standardization. 2014. SIST EN 12464-2:2014 Light and lighting - Lighting of work places - Part 2: Outdoor work places. Accessed Febraury February 13, 2024. https://standards.iteh.ai/catalog/standards/sist/b1a68cbc-1bdc-4ac2-89e7-0123982763b5/sist-en-12464-2-2014
6. International Organization for Standardization. ISO 8995-1:2002 (CIE S 008/E:2001) - Lighting of work places -- Part 1: Indoor. 2001. https://www.iso.org/standard/28857.html.
7. Zheng W. GB 50034-2013 Standard for lighting design of buildings (English Version). https://books.google.com/books/about/GB_50034_2013_Translated_English_of_Chin.html?id=svamDgAAQBAJ. Published 2013. Accessed April 1, 2018.
8. The SLL Lighting Handbook The Society of Light and Lighting. CIBSE; 2018. www.cibse.org. Accessed June 9, 2020.
9. U.S. General Services Administration. P100 Facilities Standards for the Public Buildings Service.; 2018. https://www.gsa.gov/cdnstatic/2018 P100 Final 5-7-19_0.pdf.